Keeping the House in the Family: Non-Market Transfers and Property-Tax Incentives *

Saannidhya Rawat

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Abstract

A large share of American homes change hands without a market. Using 144 million residential deed events from 2007 to 2023, this paper provides national micro-level evidence on the prevalence and size of intra-family housing transfers. We find 28 family transfers per 100 market sales, and this family channel increases with tax advantages and is associated with older housing stock. California's Proposition 19 provides evidence on how inherited property-tax assessments shape this channel: transfers surged 31% above counterfactual before the deadline and then declined. Synthetic-control specifications yield 2022–2023 gaps of 27–33%; a market-adjusted DiD corresponds to roughly 58,045 missing deeds annually. The missing deeds have not been shown to reappear as market sales. A simple framework shows that when heirs inherit a below-market property-tax assessment along with the house, as permitted by California's Propositions 13, 58, and 193, families keep homes that the market values more. Our results suggest that family transfers are not outside the housing market but are one of its allocation mechanisms and property-tax rules can shift scarce homes from price-based matching to lineage-based retention.

JEL classification: R21, R31, H22, D64

Keywords: housing supply, intergenerational transfers, bequests, property taxes, Proposition 19

*Saannidhya Rawat, Marquette University. Email: saannidhya.rawat@marquette.edu. CoreLogic deed and property records were obtained through the University of Cincinnati's academic license. All errors are our own.

1 Introduction

Housing turnover is usually measured through market sales. Families also transfer homes by gift, succession, and other non-market conveyances. These deeds can change legal title without generating a market price or changing who occupies the property. Between 2007 and 2023, our classification of CoreLogic residential records identifies nearly 20 million family-transfer deeds, one for every 3.6 market sales. Understanding this channel requires measuring both its scale and the institutions that shape it.

This paper has three parts. First, it establishes national facts about within-family transfers using 144 million residential parcel-date events. Family deeds represent a substantial and persistent flow alongside market sales. Their ratio to market sales is highest in California, where the tax code historically allowed heirs to inherit a property's assessment as well as the property. Their share rises across middle-aged housing vintages, although it is not monotone over all ages. Many properties remain unsold long after a family transfer. These patterns describe a channel of housing ownership; none by itself identifies inefficient use or a causal tax effect.

Second, the paper examines how tax incentives shape recorded family transfers using California's Proposition 19. Under Propositions 58 and 193, eligible intergenerational transfers could preserve the parent's assessed value. Proposition 19 narrowed this exclusion in February 2021, retaining conditional protection for a principal residence occupied by the heir and for family farms while removing it for other qualifying transfers (California State Board of Equalization, 2025). The LAO estimated that 60,000–80,000 properties annually had received inheritance exclusions under the earlier regime (California Legislative Analyst's Office, 2017). That is an administrative property count, not a tax-revenue estimate or a count of inefficiently allocated homes.

California family deeds surged 31 percent above the transparent counterfactual in the four months before the deadline. They then fell to a persistently lower level. A synthetic control estimates a 33 percent 2022–2023 gap; adjusting for the population aged 65 and over yields 26 percent. These are separate specifications, not the endpoints of a confidence interval. A DiD adjustment for California's market-sale decline implies roughly 58,045 missing deeds annually when scaled by the 2018–2019 baseline. Inference varies across designs: the raw SCM post/pre-MSPE rank has $p = 0.039$, while the net-DiD state-placebo rank has $p = 0.176$. Conditional sale hazards and estate/trust-seller market sales do not establish an accompanying release of homes through 2023. Deferred succession is consistent with the evidence but is not identified by those null results.

Third, a simple framework quantifies the potential allocation gains from removing the

inherited-assessment advantage. The object is the distortion created by the tax wedge, not the existence of non-market transfers. Family attachment, insurance, avoided transaction costs, and efficient rental use can make retention socially valuable. Inefficient retention arises only when the tax advantage induces a family to forgo an alternative use with greater net surplus. Recorded deeds do not directly identify that margin.

We therefore combine the estimated deed response with explicit assumptions about the conversion to distinct allocation-changing properties, the tax wedge, succession delay, and persistence. California parcel episodes supply a descriptive persistence proxy, and an LAO example for long-held Los Angeles homes supplies an external wedge benchmark. Under a uniform valuation-gap approximation, the annual gain per marginal property is half the wedge. This sufficient-statistics-inspired calculation retains the model assumptions needed for welfare interpretation (Chetty, 2009). The resulting tables show annual gains, property-years, and discounted gains, including zero when deeds change but allocations do not. They are conditional policy calculations, not estimates of realized or total Proposition 19 welfare.

The paper connects the economics of bequests and inter-vivos transfers (Bernheim et al., 1985; Kopczuk, 2007; McGarry, 1999; Piketty, 2011) to an asset-level record of housing ownership. Research on intergenerational wealth transmission (Charles and Hurst, 2003) motivates the importance of the asset, while evidence on heirs' property documents a distinct source of title-related immobility (Burnett and Winters-Michaud, 2025; Dobbs and Johnson Gaither, 2023). Recent work studies intergenerational housing-capital persistence using linked Census, property, and income-tax records (Binder et al., 2026). Our deed-based object is the conveyance of particular assets, rather than the persistence of children's housing holdings. Closely related work by Baker et al. (2026) uses parent-child exclusion claims in Los Angeles and San Francisco to study Proposition 19 retiming and subsequent sales. The present national measurement and California-wide comparisons complement that evidence; our recorder-based family proxy covers a broader population than confirmed parent-child exclusion claims. The nearest descriptive antecedents also include industry tabulations (Delventhal, 2026) and the LAO's assessment-roll accounting.

The analysis also extends the property-tax and housing lock-in literature (Coven et al., 2026; Ferreira, 2010; Fonseca and Liu, 2024; Wasi and White, 2005) to intergenerational title transfers. The transaction-tax literature (Best and Kleven, 2018; Hilber and Lyytikäinen, 2017; Kopczuk and Munroe, 2015; Slemrod et al., 2017) provides a comparison for timing and volume responses, although a deadline response alone does not reveal the valuation gaps required for welfare. Misallocation concerns, as in Glaeser and

Luttmer (2003), require evidence or assumptions about forgone surplus. Older housing can serve an important filtering role (Rosenthal, 2014), but a family-owned rental may already provide those services to a tenant.

Section 2 describes the data and classification. Section 3 presents the four national facts. Section 4 studies the policy response, and Section 5 develops the framework and conditional welfare calibration. Section 6 concludes.

2 Data and the Anatomy of a Non-Market Transfer

2.1 Deeds

Property records come from CoreLogic's Owner Transfer file (CoreLogic, 2024), obtained through an academic license, which compiles the universe of recorded deeds from county recorder offices. A deed is filed whenever the legal ownership of a parcel changes—not only when a house is sold. The same administrative trail that records a $400,000 sale records a mother quitclaiming her house to her son for nothing. Market-based datasets (MLS listings, repeat-sales indices, mortgage originations) discard these zero-price events by construction. Here they are the object of study.

The extract covers all fifty states and the District of Columbia. Coverage is dense from 2007 through mid-2024; we use 2007–2023 for annual statistics and treat 2024 as a partial year. Restricting to residential parcels and deduplicating to one event per parcel per derived sale date yields 148.7 million deed events (144.0 million in 2007–2023), linked through a permanent parcel identifier that follows each house across successive owners.

2.2 Classifying transfers

CoreLogic flags each record with a primary category—arm's length or not—and an interfamily indicator constructed from the recorder's consideration, deed type, and name fields. We sharpen these flags into a taxonomy using the buyer and seller names on the deed:

1. **Market sale**: arm's-length category, recorded price of at least $10,000, not flagged interfamily (71.1 million events, 2007–2023).

2. **Same-person retitle**: the named principal is the same person on both sides of the deed—an owner adding or removing a co-owner, potentially at marriage, divorce,

or refinancing (7.1 million). The matched name remains, but ownership shares can change. These deeds are tracked separately from the operational family aggregate.

3. **Family transfer, same surname**: flagged interfamily, buyer and seller sharing a surname but not the same person, no trust detected (13.6 million). This is a narrower proxy for inter-person family transfers: it misses unmatched surnames but can also retain same-person events with inconsistent full names. It is not a formal lower bound on true family transfers.

4. **Family transfer, other**: flagged interfamily without a detected surname match after the earlier exclusions, plus the separately coded estate/executor residual (6.1 million). Missing surname fields and transfers involving entities can enter this group; the code does not verify the relationship between the parties.

5. **Trust self-transfer (operational label)**: the interfamily flag is set and a primary party name contains a trust keyword (12.1 million). This can include estate-planning retitling and distributions to different beneficiaries. Unchanged control is not verified; the class is excluded from the family aggregate.

6. **Other non-arm's-length**: the remainder—divorce divisions across surnames, title corrections, nominal-price conveyances, and foreclosure-related instruments (34.0 million).

Table 1 reports observable differences across the classes. Same-surname family deeds differ sharply from the defined market-sales class: 90 percent of same-surname transfers record a zero or missing price, against 0 percent of market sales (which require a price by construction) and 65 percent of the residual non-arm's-length class; 40 percent travel by quitclaim deed—the instrument of gift—against 2 percent of market sales. Where a price is recorded at all in a same-surname transfer, the median is $97,000, well under half the $224,000 median market price: the comparison is descriptive because the houses and transferred interests can differ. Corporate buyers, who take 10 percent of market purchases, are essentially absent from same-surname transfers. These patterns are consistent with gifts and succession, but do not verify their prevalence or exclude other transactions. One note on probate: formal estate keywords ("estate of," executor, administrator) rarely appear in the recorded name fields, and the earlier trust, same-person, and surname rules also take precedence over the estate rule. The small residual estate class therefore cannot measure all successions.

Table 1: Residential deed events by class, 2007–2023

Event class	N (2007–2023)	Zero/no price (%)	Median price[a]	Quitclaim share (%)	Absentee recipient (%)[b]
Market sale (arm's length)	71,076,255	0.0	$224k	2.0	26.8
Other non-arm's-length	33,980,735	64.8	$57k	17.8	55.9
Family transfer: same surname	13,610,459	90.3	$97k	39.6	17.8
Trust self-transfer	12,074,757	96.1	$130k	30.0	25.6
Same-person retitle (co-owner change)	7,123,579	92.3	$63k	40.3	16.9
Family transfer: other interfamily	6,073,863	88.3	$155k	26.4	34.8
Estate deed (non-interfamily)	57,697	9.6	$167k	2.2	89.8
Family transfer: estate/executor	1,905	82.9	$596k	8.6	35.6

Notes: Authors' classification of CoreLogic Owner Transfer records, residential parcels, deduplicated to one event per parcel-date. [a]Median of strictly positive recorded prices. [b]Share of events whose buyer mailing ZIP differs from the property ZIP, among events where both are observed.

3 Within-Family Housing Transfers: National Facts

Fact 1: For every 3.6 market sales, one family-transfer deed is recorded. Table 2 reports annual volumes; Figure 1 shows the pooled scale and annual persistence of the family channel. Between 0.9 and 1.6 million family-transfer deeds are recorded every year (same-surname, other-interfamily, and estate/executor), against 3.0 to 5.8 million market sales: a family:market ratio between 0.25 and 0.34 in every year of the sample, with a 2007–2023 aggregate of 0.28. The conservative same-surname series alone runs 0.7 to 1.0 million per year. The channel is not a recession artifact: the ratio was as high in the 2009 bust (0.29) as in the 2021 boom (0.28). And the count is deliberately strict—it excludes the 12.1 million trust self-transfers, 7.1 million same-person retitles, and 34.0 million other non-market conveyances; adding them, the deed traffic outside the market rivals the market itself.[1]

Fact 2: Family transfers are concentrated in California. Figure 2 ranks states by the family:market ratio over 2017–2023. California leads the nation at 0.61—one family transfer for every 1.6 market sales, 2.3 times the pooled national ratio of 0.26—ahead of Utah and Idaho (0.53). California is exactly where acquisition-value assessment made the inherited tax basis most valuable, and it also accounts for 38 percent of all trust self-transfers in the country (2.2 million over seven years, more than its family transfers), consistent with extensive estate-planning activity. Cross-state correlations are

[1]Restricting the denominator to existing homes (excluding new-construction first sales) raises the aggregate family:market ratio to 0.30. On a per-property rather than per-deed basis, the 19.7 million family-transfer deeds cover 16.5 million distinct parcels (1.19 deeds per parcel).

Table 2: The parallel housing market: annual volumes

Year	Market sales	Family (same surname)	Family (broad)	Trust self-transfers	Family:market ratio (broad)
2007	3,906,268	884,266	1,068,273	518,750	0.27
2010	3,097,287	723,862	959,657	592,088	0.31
2013	3,793,643	803,373	1,132,061	680,645	0.30
2016	4,565,213	804,689	1,178,714	739,483	0.26
2019	4,956,337	819,739	1,315,281	741,526	0.27
2021	5,820,867	1,047,039	1,622,749	966,149	0.28
2022	5,076,718	792,479	1,284,736	901,638	0.25
2023	4,113,610	653,050	1,061,180	951,273	0.26

Notes: Residential deed events. "Family (broad)" = same-surname + other-interfamily + estate/executor classes; excludes trust self-transfers. Ratio = family (broad) / market sales.

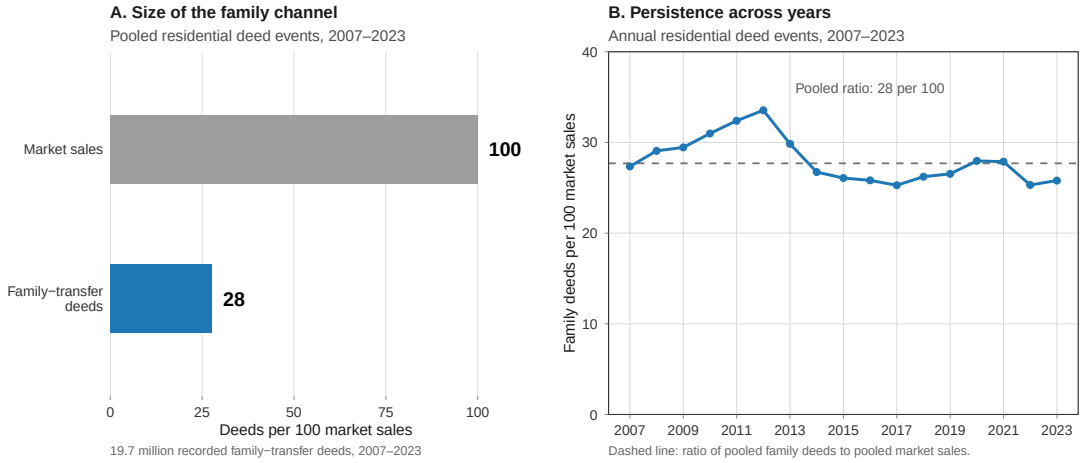


Figure 1: Size and persistence of the family-transfer channel, 2007–2023. Panel A normalizes pooled market sales to 100 and shows approximately 28 family-transfer deeds per 100 market sales. Panel B reports annual family-transfer deeds per 100 market sales; the dashed line is the pooled ratio (27.7). Family transfer (broad) combines same-surname, other-interfamily, and estate/executor classes and excludes trust self-transfers. Counts refer to deed events, not distinct properties.

not causal evidence—demography, land tenure, and recording practice all differ—but the cross-section puts the tax-subsidy hypothesis on the table, and Section 4 tests it within California.

Fact 3: Family transfer shares are higher across middle-aged housing vintages. Linking deed events to property characteristics (61 percent of 2017–2023 selected family/trust/market events match the available property-characteristics snapshot;

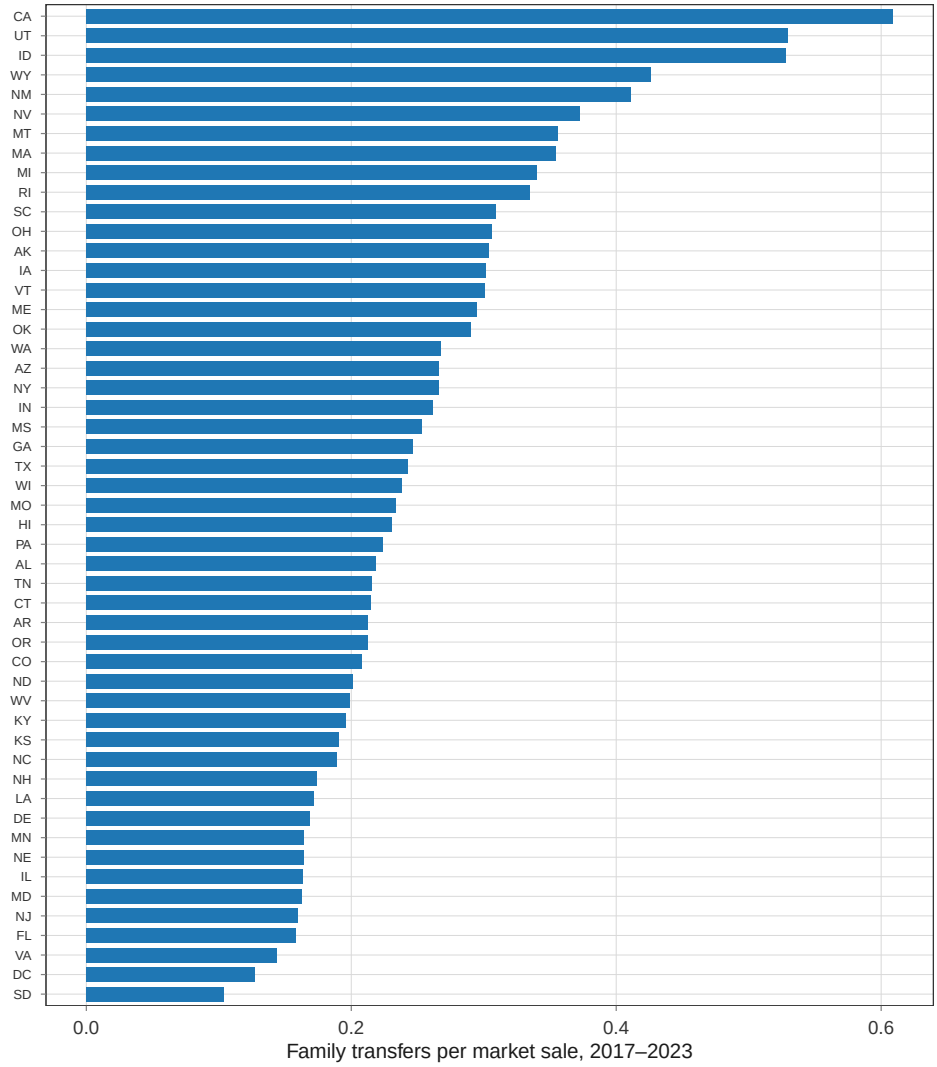


Figure 2: Family transfers (broad) per market sale by state, 2017–2023.

Appendix A), the family share of within-class traffic rises with structure age: family transfers are 20 percent of combined family-plus-market events for houses of ten to thirty years, 22 percent at thirty to fifty years, and 26 percent at fifty to seventy-five years, before falling to 24 percent for homes aged 75 or more. The gradient is therefore not monotone over all vintages; the under-ten-year bin also has very few matched events. The family share is higher on parcels with senior tax exemptions (26 versus 22 percent)—an association measured with a historical snapshot—and declines monotonically across the within-state value distribution, from 25 percent in the bottom quintile to 21 percent in the top. The houses that families keep are disproportionately the older, cheaper stock through which housing filters down to lower-income buyers (Rosenthal, 2014).

Property types distinguish different family channels. Using recorded land-use codes, family deeds account for 26.6 percent of combined family-plus-market deeds for duplexes, 23.0 percent for single-family properties, 15.0 percent for condominiums, and 13.0 percent for townhouses or rowhouses. On common state-year support, weighting each type to the single-family distribution leaves the duplex share 2.1 percentage points higher and the condominium and townhouse shares 8.3 and 6.7 points lower, respectively. Appendix A.1 gives the definitions and denominators. These are static recorded types, not measures of occupancy or rental use; the apartment sample is selectively covered. The variation motivates distinguishing inheritance of a residence from inheritance of housing capital without treating building type as evidence of either motive.

Fact 4: Three in five family-transferred homes are still unsold a decade later. What happens to a house after it enters the family channel? Because the parcel identifier links successive deeds, every transfer event can be followed to the property's next arm's-length market sale, if any. We take all events in the 2008–2018 cohorts—guaranteeing at least 66 months of follow-up before the June 2024 censoring date—and estimate Kaplan–Meier survival to first market sale, separately by event class: 8.5 million same-surname family transfers, 3.4 million other family transfers, 7.1 million trust self-transfers, 4.1 million same-person retitles, and 42.1 million market purchases as the benchmark.

Table 3 and Figure 3 show two distinct family destinies. The first is settlement: among cross-surname family conveyances—the class containing estate distributions among multiple heirs—10 percent reach the open market within twelve months and 15 percent within twenty-four, roughly double the market-purchase benchmark (5 and 9 percent). For a substantial minority, the family deed is the administrative final step before the house is sold.

The second destiny is the lock. Once a family-transferred home survives the settlement window, it becomes more immobile than the housing stock at large. Same-surname transfers start faster than market purchases (11.1 versus 8.9percent sold within two years) but fall behind by year five (23.1 versus 23.7 percent) and keep falling behind: by year ten, 38 percent of same-surname family homes have returned to the market against 42 percent of market purchases. Conditional on passing the two-year settlement phase, 30 percent of family homes sell over the following eight years versus 36 percent of market purchases. The market-purchase benchmark, recall, includes investors and flippers; family homes are stickier than all of it. Three in five of the homes that passed between relatives in 2008–2018 had not sold on the open market a decade later. An internal descriptive

Table 3: Time to next open-market sale, 2008–2018 event cohorts

		Sold on open market within (%)			
Event class	N	12m	24m	36m	60m
Market sale (arm's length)	42,055,369	5.2	8.9	13.7	23.7
Family transfer: same surname	8,467,604	6.9	11.1	15.4	23.1
Family transfer: other interfamily	3,426,450	10.1	14.7	18.9	26.5
Family transfer: estate/executor	665	9.9	14.3	41.2	45.9
Trust self-transfer	7,107,100	4.7	8.9	13.1	21.0

Notes: Share of events followed by an arm's-length market sale of the same parcel within the stated horizon. All cohorts observed at least 66 months before the June 2024 censoring date.

check is that same-person retitles—deeds retaining at least one matched name—track trust self-transfers closely (9.1 versus 8.9 percent sold within two years), consistent with both classes often recording paperwork rather than a change in the property's use.

This is a descriptive fact, and selection is its content: families keep the homes they receive. Four caveats discipline its reading. First, the comparison pools states and cohorts as family transfers concentrate in older homes and in low-turnover states, so part of the gap is composition. Second, the two clocks measure different objects: a market purchase starts a fresh tenure, while a family deed often changes title without changing occupancy—a force that should, if anything, *raise* the family class's sale hazard, if family deeds cluster near succession events. Third, some cross-surname family homes cannot sell for title reasons (heirs' property), a lock of a different kind (Dobbs and Johnson Gaither, 2023). Fourth, trust self-transfers —the trust-keyword class—show survival close to the same-surname class, a reminder that long-tenure households are exactly the ones that do estate paperwork. These comparisons do not distinguish attachment, rental use, title constraints, and tax incentives. Proposition 19 identifies a response of deeds to a policy change; additional assumptions are needed to connect that response to allocation.

4 Tax Incentives & Family Transfers: Proposition 19

4.1 Institutional background

Proposition 13 (1978) fixed each California parcel's assessed value at its acquisition price, with growth capped at two percent per year; reassessment to market value occurs only at a change in ownership. Proposition 58 (1986) exempted parent-to-child transfers from this reassessment—for a principal residence of unlimited value plus up to $1 million of

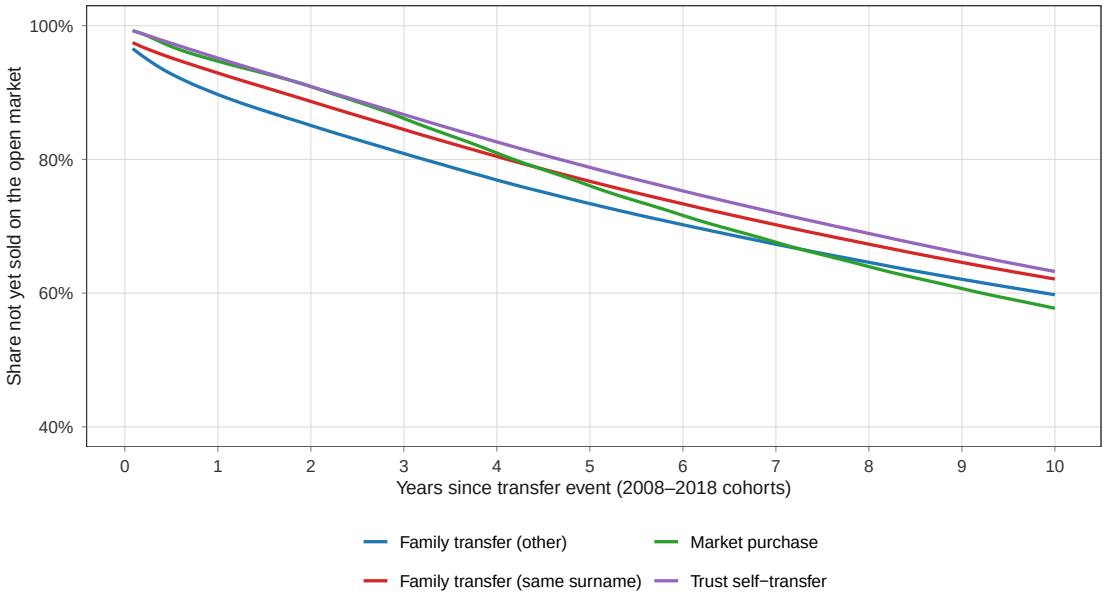


Figure 3: Kaplan–Meier survival to first open-market sale, 2008–2018 event cohorts, censored June 2024. The estate/executor class (n=665) is omitted.

assessed value of other property—and Proposition 193 (1996) extended the exemption to grandparent-grandchild transfers. A child could therefore inherit not just a house but its decades-old tax bill, and keep that tax bill while renting the house out. The Legislative Analyst's Office estimated that 60,000 to 80,000 inherited properties escaped reassessment each year under this regime (California Legislative Analyst's Office, 2017).

Proposition 19, passed by voters on November 3, 2020, narrowed the exclusion sharply, effective February 16, 2021: it now applies only when the home was the parent's principal residence *and* the child makes it their own principal residence within a year, and only up to the inherited taxable value plus $1 million (inflation-adjusted); the exclusion for rentals, vacation homes, and other non-primary-residence property was eliminated, with a carve-out for family farms (California Legislative Analyst's Office, 2020; California State Board of Equalization, 2025). For an otherwise eligible non-occupant heir, the reform removes the inherited-assessment advantage. This raises the tax cost of retention relative to sale, although the family may respond by changing paperwork, timing, occupancy, or ownership rather than by listing the home. County assessors reported a flood of parent-child filings ahead of the deadline.[2] The reform's other arm—expanded portability of

[2]Sonoma County, for example, received roughly five times its typical volume of parent-child exclusion filings in the weeks before the February 15, 2021 deadline (North Bay Business Journal, February 2021); the Los Angeles County assessor accumulated a multi-year backlog of roughly 20,000 claims (Los Angeles County Auditor-Controller, 2025).

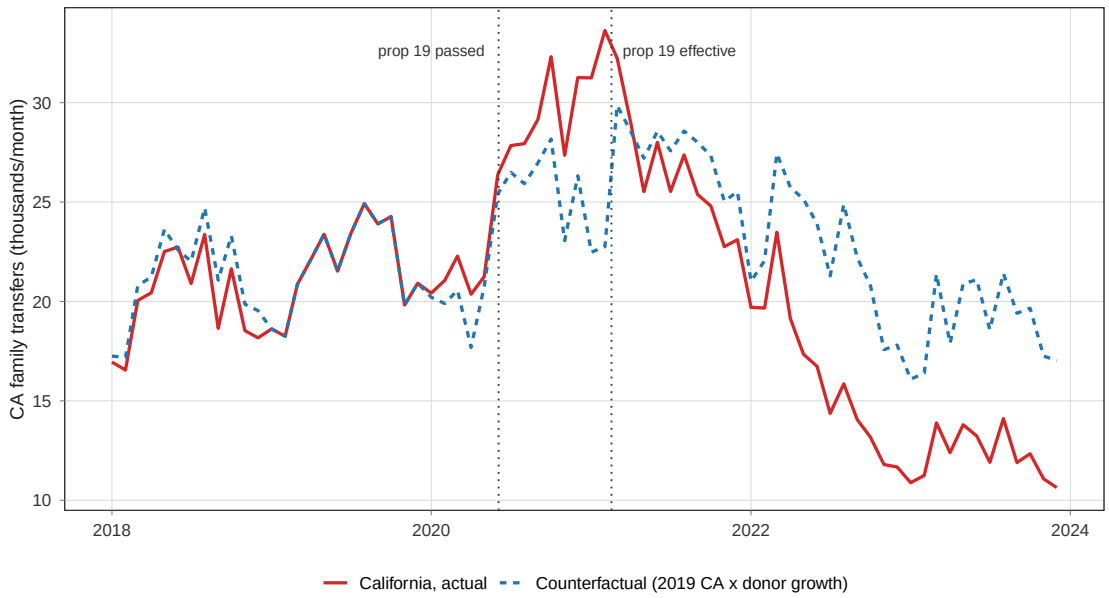


Figure 4: California monthly family transfers (broad class) versus counterfactual (CA 2019 same-calendar-month volume × donor-state growth). Dotted lines: November 3, 2020 (passage) and February 16, 2021 (parent-child provisions effective).

assessments for movers aged 55 and over, effective April 1, 2021—affects market sales rather than family transfers; the market-sale placebo therefore need not be an untreated control. We report its adjustment separately from the synthetic-control estimates.

Throughout this section, "family transfers" means the broad class (same-surname, cross-surname, and estate/executor deeds), measured monthly by state from the event panel; results including trust self-transfers are reported as robustness.

Figure 4 plots monthly California family transfers against a transparent counterfactual: California's 2019 volume in the same calendar month, scaled by contemporaneous family-transfer growth in the other 49 states and DC. The counterfactual thus inherits both California's seasonality and national shocks—COVID disruption, the 2020–2021 housing boom, the estate-planning surge—while excluding California from the donor pool. Before passage, it fits tightly: over 2018m1–2020m10, the mean absolute monthly error is 3.9 percent and the RMSPE is 5.7 percent.

California tracks the counterfactual closely until November 2020. Between passage and the deadline, the two series tear apart: from November 2020 through February 2021, actual family transfers total 123,486 against a counterfactual of 94,615—an excess of 28,871 deeds, 31 percent above counterfactual, crammed into four months. February 2021, ordinarily the year's quietest month, became the busiest family-transfer month in

the 2018–2023 California series. This is the anticipatory bunching that notch designs recover for taxed market transactions (Best and Kleven, 2018; Slemrod et al., 2017), reproduced in transfers with no price: families re-timed successions to bequeath a tax basis. In leave-one-out placebo counterfactuals, California's anticipation gap is third of fifty-one in raw size ($p = 0.059$) but first after dividing each state's gap by its own pre-period RMSPE ($p = 0.020$). The concentration around the deadline supports a tax-timing interpretation, although monthly counts do not identify the exact legal transfer date or rule out every contemporaneous influence.

4.2 Empirical Strategy

We compare California with the other 49 states and the District of Columbia. Identification requires that, absent the reform, California's family-transfer trajectory would have followed the relevant control trajectory.

4.2.1 Synthetic control

Following Abadie et al. (2010), we construct a convex combination of donor states using monthly family-transfer counts normalized by each state's 2018–2019 monthly mean. We choose weights to match the January 2018–May 2020 trajectory, ending the fitting window before the June 2020 ballot announcement. The post-reform effect is the average proportional gap relative to synthetic California in 2022–2023. Inference uses in-space placebos ranked by the post/pre mean squared prediction error ratio and intervals based on donor-state gaps. As a demographic sensitivity check, we normalize transfers by the population aged 65 and over; this specification fits through October 2020.

4.2.2 Difference-in-differences

Using annual data, we estimate the following difference-in-differences specification:

$$\log F_{sy} = \alpha_s + \lambda_y + \delta \, \mathbf{1}\{s = \text{CA}\}\mathbf{1}\{y \geq 2022\} + \varepsilon_{sy}, \tag{1}$$

where F_{sy} denotes family transfers in state s and year y. The sample includes 2017–2019 and 2022–2023, excluding the anticipation and transition years 2020–2021. We estimate the same specification for market sales and report the difference between the two coefficients separately, since assessment portability may also affect market sales. Inference reassigns treatment across states and ranks California's absolute coefficient in the resulting

placebo distribution.

4.2.3 Event-study design

To separate anticipation from the post-deadline response, we estimate a monthly event study over 2018–2023 using the following specification:

$$\log F_{st} = \alpha_s + \lambda_t + \sum_{\tau \in \mathcal{T} \setminus \mathcal{B}} \beta_\tau \, \mathbf{1}\{s = \text{CA}\}\mathbf{1}\{t = \tau\} + \varepsilon_{st}, \tag{2}$$

where t indexes calendar months, $\mathcal{T}$ is the sample period, and $\mathcal{B}$ pools January 2018–December 2019 as the omitted baseline. State and calendar-month fixed effects absorb persistent state differences and common shocks; β_τ measures California's differential change relative to the baseline. We report clustered confidence intervals at the state level.

4.3 Results: The Rush and the post-reform decline

Retiming alone would imply a transitory post-deadline trough followed by recovery. Instead, the gap persists and widens. From March 2021 through December 2022, actual transfers run 85,351 below counterfactual (16 percent); by 2023, California family transfers (147,441) are 41 percent below their 2018–2019 average (251,256)—while national volumes returned to trend. A donor-weighted synthetic control sharpens the comparison. Following Abadie et al. (2010), we build a synthetic California from a convex combination of the other 49 states and the District of Columbia, with weights chosen to match California's pre-reform family-transfer trajectory. Because California is the largest state—an outlier in the *level* of transfers, which forces a level-matched synthetic onto a single large donor—we match on each state's volume normalized to its own 2018–2019 mean. The synthetic reproduces California's pre-reform path to a root mean squared error of 5.0 percent (Figure 5); after the February 2021 deadline, California's family transfers run 33 percent below synthetic California through 2022–2023.

The difference-in-differences estimates in Table 4 formalize this. Using state-by-year counts for 2017–2019 versus 2022–2023 (the transition years 2020–2021 are excluded as contaminated by anticipation and retiming), with state and year fixed effects, log family transfers fall by 0.425 (s.e. 0.035) in California relative to other states—35 percent. The placebo column shows log market sales falling 0.163 (15 percent) over the same window, California's general post-rate-shock slump; netting it out leaves a 23 percent decline specific to family transfers, roughly 58,000 transfers per year against the 2018–2019 base.

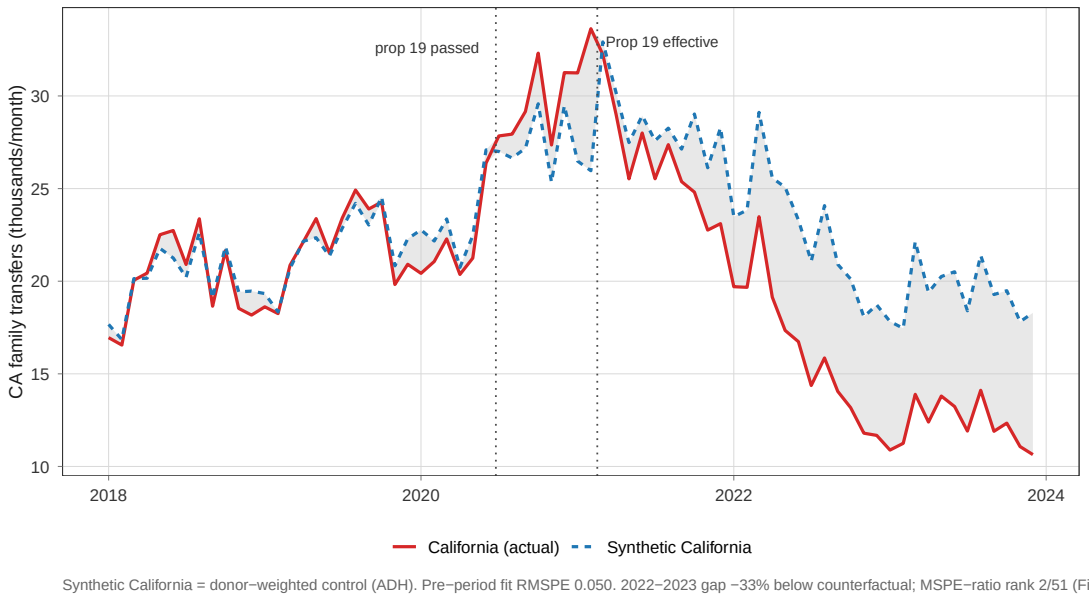


Figure 5: California monthly family transfers against a donor-weighted synthetic California (normalized-index match; Abadie et al., 2010). The series track closely before the reform and diverge after the deadline. Dotted lines: passage (November 3, 2020) and the parent-child effective date (February 16, 2021).

Reclassification into trusts cannot explain it: including trust self-transfers in the family aggregate yields a decline of 0.241 log points (21 percent).

Figure 6 shows monthly event-study results. We regress log family transfers on California-by-month indicators with state and year-month fixed effects, the other 49 states and DC as controls, and the 2018–2019 months pooled as the omitted baseline. California sits near its 2018–2019 baseline in early 2020, drifts upward through the anticipation period as Proposition 19 reaches the ballot and passes, spikes to +0.32 log points in February 2021—the deadline month—and then settles into a persistent decline of 0.3 to 0.5 log points through 2022–2023. An annual event study reports essentially zero for 2021 only because the pre-deadline rush and the post-deadline collapse cancel within the calendar year; at monthly resolution the two are plainly distinct, and the post-reform level shift is unmistakable. The missing annual flow, roughly 58,045 deeds, is of a similar order to the Legislative Analyst's 60,000–80,000 count of tax-excluded inherited properties per year. The two quantities cover different populations and units; this comparison does not validate the estimated deed response or identify how many properties change allocation.

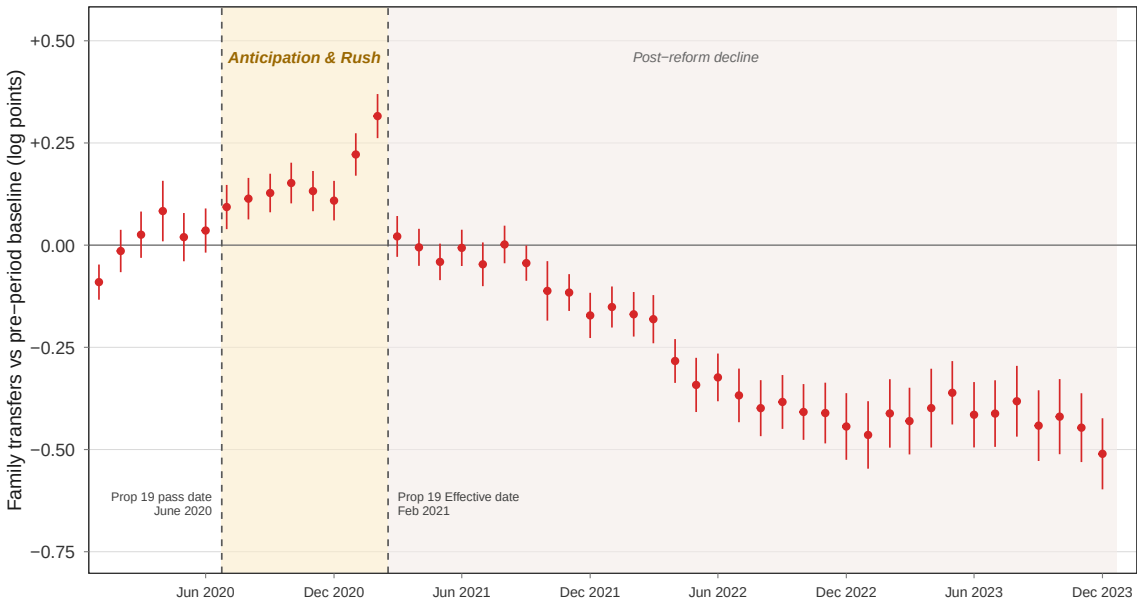


Figure 6: Monthly event study of California family transfers. Points are California-by-month coefficients from a regression of log family transfers on state and year-month fixed effects, with the 2018–2019 months pooled as the omitted baseline and the other 49 states and DC as controls. Whiskers are 95 percent state-clustered confidence intervals. Shaded regions mark the anticipation build-up (ballot to passage), the rush between passage and the February 16, 2021 effective date, and the post-reform period; dashed lines mark the passage and effective dates.

4.3.1 Selection: who still keeps the house

If the reform discourages transfers to heirs who would otherwise retain the house for its tax basis, the surviving transfers can become more selected toward long-term retention. This is a possible mechanism, formalized in Section 5, rather than an implication of the volume estimate alone. We take family-transfer cohorts from two clean windows—January 2017 to December 2018 (pre) and July 2021 to June 2022 (post, after the retiming turbulence)—and measure whether each transferred home reaches the open market within 24 months, a window that closes before the June 2024 censor for every cohort. Corporate recipients are excluded. The design compares California to other states, cohort month by cohort month (state and cohort-month fixed effects, weighted by cell size, clustered by state), with the identical regression run on market *purchases* as a placebo for California-specific housing-cycle shocks; the triple-difference nets the two.

The raw numbers tell the story (Appendix Figure A4). Before the reform, California family-transferred homes reached the market *more* often than the national average (16.1 versus 13.1 percent within 24 months)—California's inherited-basis regime made even

Table 4: Proposition 19: volume, selection, and composition

	Log volume		Sold within 24 months		Absentee recipient
	Family transfers	Market sales	Family transfers	Market purchases	Family transfers
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
CA × Post	-0.4253	-0.1626	-0.0178	-0.0104	-0.0017
	(0.0352)	(0.0293)	(0.0032)	(0.0019)	(0.0032)
Fixed-effects					
State	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes			
Cohort month			Yes	Yes	Yes
Fit statistics					
Observations	255	255	1,836	1,836	1,836
R^2	0.98575	0.98961	0.84935	0.85148	0.90779

Clustered (State) standard-errors in parentheses

Columns (1)-(2): state-year counts, 2017-2019 versus 2022-2023. Columns (3)-(5): transfer-cohort cells, January 2017-December 2018 versus July 2021-June 2022, weighted by cell size. Standard errors are clustered by state and shown for scale; inference with one treated state uses state placebos.

hold-motivated transfers cheap to unwind later, and its hot market made selling attractive. After the reform, California converges toward the national rate (12.3 versus 11.1 percent). In the regression, the California post-reform effect is −1.78 percentage points; the market-purchase placebo shows −1.04 points, and the triple-difference is −0.74 points—a decline, as the selection logic predicts, not a rise. But candor about inference is required here: the triple-difference sits well within the placebo-state distribution (permutation $p = 0.67$), and the absentee-recipient share is flat (−0.2 percentage points). These imprecise conditional effects do not independently identify selection or rule out a change in settlement deeds. They concern the transfers still observed after the reform, not the missing transfers. The clearest response is in recorded volume.

4.3.2 Do missing deeds reappear as estate sales?

If the reform redirected near-succession transfers into market sales, one possible implication would be more sales by estates and trusts. Advance tax-planning deeds need not produce this pattern immediately. We examine market sales whose seller carries an estate, executor, or trust designation, while recognizing that heirs' later sales need not retain those keywords. The test does not detect a statistically clear increase. Estate- and trust-seller market sales in California fall 8 percent relative to donor states between 2017–2019 and 2022–2023 (permutation $p = 0.71$); relative to California's overall market slump they *gain* 8 log points—their share of California market sales rises from 28.4 to 31.9 percent—but similar share gains occur elsewhere (permutation $p = 0.63$).

The absence of a statistically clear increase is consistent with deferred succession, but it does not establish that the missing deeds were predominantly inter-vivos transfers. Limited power, changes in recording or legal vehicles, and later sales by heirs without estate keywords can also weaken this test. The data therefore do not yet identify a realized supply release or guarantee one in the future. Section 5 treats the conversion of missing deeds into allocation changes, and the delay before those changes, as explicit calibration assumptions.

5 Family Channel: A Simple Framework

The policy experiment identifies a response of recorded deeds. Translating that response into welfare requires two further steps: determining which properties change allocation, and valuing those changes. This section makes those steps explicit. Its quantitative results are conditional calibrations of potential gains from the inheritance provisions, not estimates of realized gains or the total welfare effect of Proposition 19.

5.1 Framework

Consider a price-taking marginal market buyer and a marginal heir. Let u_m be the flow per-period value of the house to the marginal market buyer and u_h be the value of the house to the heir, including attachment and the net value of rental use.

Suppose the property-tax rate is t, assessments reset to the sale price when a property is sold, and the discount rate is r. Capitalizing the perpetual tax burden, the marginal market price is

$$p = \frac{u_m}{r + t}. \tag{3}$$

Now consider an heir at the moment of succession who values living in the house, or renting it out, at flow u_h. If the family keeps the house, the heir pays taxes on assessed value A. If the family sells, the heir receives $\$p$ and the house exits the family channel. Since the heir's inherited assessment is A, keeping the house is optimal when

$$\frac{u_h - tA}{r} \geq p \quad \Longleftrightarrow \quad u_h \geq u_m - t(p - A). \tag{4}$$

Equation (4) contains the framework's key trade-off. If succession triggers reassessment ($A = p$), the condition collapses to $u_h \geq u_m$: the house stays in the family only when

the heir genuinely values it more than the market does, and the family channel is allocatively harmless. But if the heir inherits the parent's assessment ($A = A_0 < p$), the family keeps the house whenever $u_h \geq u_m - t(p - A_0)$. Every heir with $u_h \in [u_m - t(p - A_0), u_m)$ retains a house that the market values more than she does. Figure 7 illustrates this interval. The wedge $t(p - A_0)$, the annual tax saving from dynastic holding, is the extra benefit the tax policy pays families to keep the houses in the family channel that would otherwise be sold in the housing market.

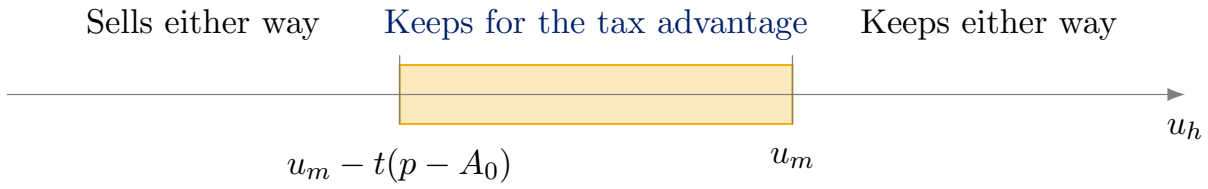


Figure 7: Tax-induced retention. With inherited assessment $A_0 < p$, heirs in the shaded interval $[u_m - t(p - A_0), u_m)$ keep the house despite valuing it less than the market buyer.

In our setup, the succession is a joint decision of the parent and the heir, and the recorded family transfer is the outcome of the keep-or-sell choice. When the family keeps, the county records a family deed; when it cashes out, the family sells directly to a market buyer and no family transfer ever appears[3]. The heir's decision is therefore a function of the tax wedge, the market price, and her own valuation. Thus, making the deed transfer decision endogenous to the tax regime. Introducing a polify reform that removes the inherited assessment ($A_0 \to p$), the following three predictions follow from the framework. First, retiming: because a family transfer must be completed before the deadline to preserve the tax advantage, the families rush to lock in A_0. Families with latent transfer plans and valuation within $[u_m - t(p - A_0), u_m)$ rush to record deeds before the effective date, and recorded transfers collapse afterward (cf. the transaction-tax notch literature: Best and Kleven, 2018; Kopczuk and Munroe, 2015; Slemrod et al., 2017). Second, the extensive margin: with the wedge removed, every heir still facing the sucession decision and with valuations in $[u_m - t(p - A_0), u_m)$ switches out of the recorded family-transfer channel. The steady-state volume of family transfers falls by the mass of heirs remaining in the wedge, and the reform potentially releases the homes to the market. Third, selection: the family transfers that survive are those with $u_h \geq u_m$—heirs who genuinely want the house, disproportionately owner-occupants, whom the surviving exclusion still favors.

[3]This simple framework abstracts from intermediary ownership structures and multiple deeds for the same property. An extension could allow a family to transfer the property to a trust before it reaches the market, or could model successive recorded deeds within a single allocation change.

The framework therefore predicts that post-reform family transfers should show a lower subsequent market-sale hazard and a lower absentee-recipient share—not because the reform failed, but because the marginal would-have-been-landlord heirs no longer take the title.

5.2 Estimating The Tax Wedge

Conditional on a common wedge w and a uniform distribution of g among switchers on $(0, w]$, the mean annual surplus gain is

$$E[g \mid \text{allocation switcher}, w] = \frac{1}{w}\int_0^w g\, dg = \frac{w}{2}. \tag{5}$$

This is the Harberger-triangle benchmark. More generally write the mean gain as λw. We report $\lambda = 0.25, 0.5, 0.75$; the conditional support restriction is $0 \leq \lambda \leq 1$. Uniformity across the entire removed wedge is an assumption about a finite reform, not a consequence of deadline bunching. Sufficient-statistics reasoning reduces the required structure but does not remove the need to justify the mapping from behavior to welfare (Chetty, 2009).

For heterogeneous properties the relevant object is the survivor-weighted sum of g_i, not the product of an unconditional average wedge and an average survival rate. Our separable benchmark holds the annual wedge and mean gain fixed and uses a common persistence profile. Sensitivities vary each component; they do not identify the joint distribution of wedge, valuation gap, and duration.

We use an external wedge benchmark because deed prices do not value unsold homes and the available property-characteristics snapshot does not identify the pre-reform assessment gap of marginal heirs. The LAO reports a \$3,000–\$4,000 annual tax saving for a long-held Los Angeles home (California Legislative Analyst's Office, 2017). Treating the unspecified dollar vintage as the 2017 publication year and using annual CPI-U gives an inflation factor of 1.2431: a \$3,729–\$4,972 range and a \$4,351 midpoint in 2023 dollars (U.S. Bureau of Labor Statistics, 2026). This is a location- and tenure-specific benchmark, not a measured statewide marginal wedge. Appendix C reports the valuation audit and half/double-midpoint stress cases.

Table 5: Inputs to the welfare calibration

Input	Value	Evidence status
Recorded-deed flow	58,045 per year	Net DiD scaled by 2018–2019 CA baseline
Net DiD inference	$p = 0.176$	State placebo rank; not a welfare test
Annual tax wedge	\$3,729–\$4,972	LAO long-held Los Angeles benchmark
Source dollar year	2017	Assumed publication-year dollars; CPI to 2023
Allocation conversion	$\alpha \in [0, 1]$	Assumed properties per missing deed
Mean surplus gap	λw	Uniform benchmark $\lambda = 1/2$
Persistence	CA parcel-episode survival	Proxy; marginal survival is not identified
Delay / horizon	0, 5, 10 / 10 years	Assumptions; grandfathered stock excluded

Notes: The tax wedge is an external benchmark, not a statewide marginal estimate. The SCM estimates are reported separately and do not enter the net-DiD reference flow. The conversion parameter includes legal-title-only responses and multiple deeds per property; it is not estimated using the national deeds-per-parcel ratio.

5.3 Estimation Results

5.3.1 Tax Wedge Distribution and Dead Weight Loss

5.3.2 Framework Results vs Reduced-Form Estimates

5.3.3 From a flow of deeds to potential property-years of gain

Let $S(a)$ describe how long the counterfactual allocation loss persists after succession, and let L be the delay between a missing deed and that succession. For a cohort c, use $F_c = F$ as an explicit constant-flow extrapolation. The potential stock of avoided misallocation at year T is

$$N_T(\alpha, L) = \sum_{c=2022}^{T} \alpha F_c \mathbf{1}\{T - c \geq L\} S(T - c - L), \qquad G_T = \lambda w N_T. \tag{6}$$

The illustration runs from 2022 through 2031. It starts with no inherited stock of reform gains: transfers already protected under Proposition 58 were not retroactively reassessed (California State Board of Equalization, 2025). Anticipatory deeds completed under the old rules are not newly treated cohorts. Continuing the 2022–2023 response for a decade is a policy scenario, not a forecast.

For persistence we construct California parcel episodes beginning in 2008–2018. An

episode begins at the first qualifying family deed since the previous market sale and ends at the next market sale or the June 30, 2024 censor. Further family deeds within the episode do not create additional houses. Ten-year survival is 59.4 percent for broad family episodes and 60.1 percent for same-surname episodes. These are descriptive proxies for persistence, not survival estimates for the unobserved marginal homes. A home can remain unsold while its use improves; conversely, an allocation loss need not end exactly at sale. Appendix sensitivities use both episode definitions and multiply cumulative hazards by one-half and two.

Table 6: Potential allocation gains: conditional calibration

α	Delay (years)	Stock in 2031 (thousands)	Property-years (thousands)	Annual gain ($m, 2031)	Ten-year PV ($m)
0.00	0	0.0	0.0	0.0	0.0
0.00	5	0.0	0.0	0.0	0.0
0.00	10	0.0	0.0	0.0	0.0
0.25	0	110.6	654.9	240.6	1,167.0
0.25	5	62.2	194.8	135.3	328.7
0.25	10	0.0	0.0	0.0	0.0
0.50	0	221.2	1,309.7	481.2	2,334.1
0.50	5	124.4	389.5	270.6	657.4
0.50	10	0.0	0.0	0.0	0.0
0.75	0	331.8	1,964.6	721.7	3,501.1
0.75	5	186.6	584.3	405.9	986.1
0.75	10	0.0	0.0	0.0	0.0
1.00	0	442.4	2,619.5	962.3	4,668.2
1.00	5	248.8	779.1	541.1	1,314.8
1.00	10	0.0	0.0	0.0	0.0

Notes: All dollars are 2023 dollars. The assumed annual wedge is the CPI-adjusted LAO midpoint; $\lambda = 1/2$, the real discount rate is 3%, and persistence uses California broad-family parcel episodes. Cohorts enter annually in 2022–2031. No value of α is empirically preferred. Rows are scenarios, not confidence intervals; zero response and ten-year delay give zero gains within the horizon.

Table 6 displays how strongly the answer depends on the unidentified conversion and delay. For illustration, $\alpha = 0.5$ and a five-year delay imply 124,378 potentially affected properties in 2031, an annual gain of $270.6 million in that year, and a discounted ten-year gain of $657.4 million. These numbers use the midpoint wedge, $\lambda = 1/2$, and a 3 percent real discount rate. They are not preferred estimates. With $\alpha = 0$, every missing deed is consistent with zero allocation gain; with a ten-year delay, no gains occur inside the reporting horizon. No positive welfare lower bound follows from the reduced-form estimates.

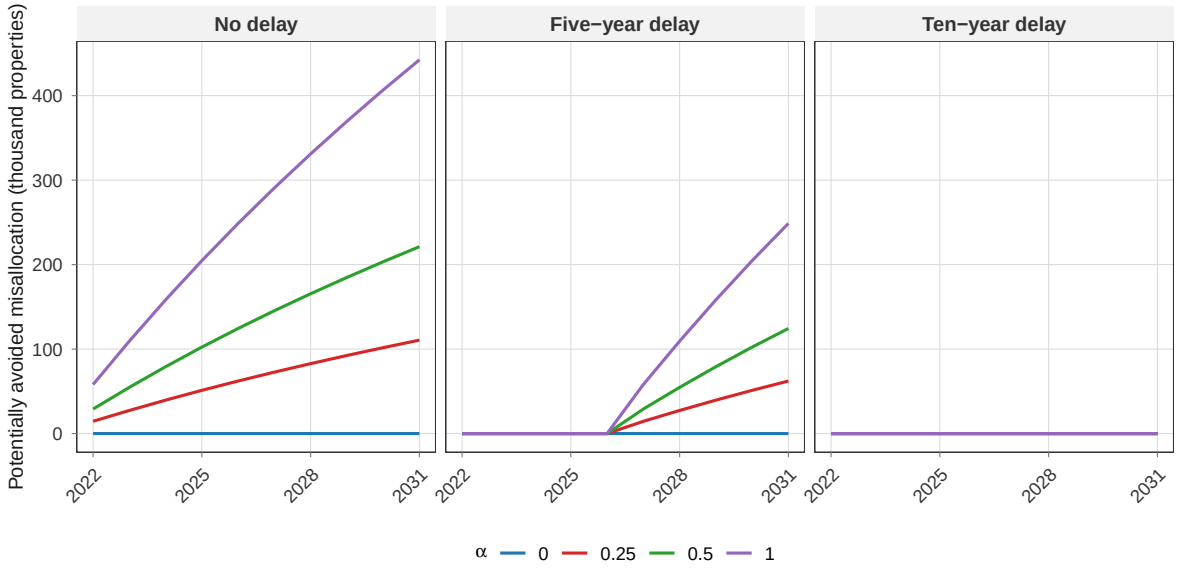


Figure 8: Potential stock of avoided misallocation under alternative conversion rates and succession delays. All paths assume a constant annual reference flow and California broad-family episode persistence. These are conditional scenarios, not observed stocks or forecasts.

5.3.4 Incidence and the limits of the policy calculation

Removing an inheritance exclusion can raise an heir's taxes and government receipts even when housing use does not change. With equal welfare weights and a dollar-for-dollar budget offset, these payments cancel in aggregate; they are not resource gains. Inframarginal heirs can bear substantial tax increases while contributing zero to G_T. Buyers and tenants gain only to the extent that housing is allocated to a more valuable use after accounting for attachment, displacement, and real switching costs.

The ledger in Appendix C separates these components. We do not estimate public-spending benefits, marginal costs of public funds, distributional welfare weights, general-equilibrium prices or rents, or gains from expanded assessment portability. Accordingly, this exercise quantifies a potential allocation component of the inheritance reform, conditional on the reduced-form interpretation and additional assumptions. It neither measures total Proposition 19 welfare nor extrapolates California's tax wedge to the national family-transfer channel.

6 Conclusion

Family transfers are a substantial part of recorded residential turnover: nearly 20 million deeds from 2007 through 2023, or one for every 3.6 market sales. Their incidence differs across states and housing vintages, and many transferred properties remain unsold for years. These facts establish the scale and persistence of a channel that price-based transaction data often omit. They do not establish that family ownership is inefficient.

California's Proposition 19 shows that this channel responds to tax incentives. The anticipation surge and subsequent decline in family deeds are consistent with a substantial role for inherited assessments in transfer decisions. The evidence on later market sales is less decisive: there is no statistically clear increase in estate/trust-seller sales through 2023, and the conditional sale-hazard estimates are imprecise. Deferral is one explanation, but altered recording and other responses cannot be excluded by those null results.

The framework separates efficient family retention from retention induced by a tax advantage. Under explicit assumptions about allocation response, tax wedges, succession delays, and persistence, the calibration translates the deed response into potential property-years and dollars of allocation gain. It also shows why the same deed response is compatible with zero gains: legal ownership can change without housing use changing, and an inherited rental can continue to serve a tenant efficiently. Tax revenue and housing surplus are separate accounts.

Longer follow-up and transaction-time information on exclusions, occupancy, rentals, and assessments could identify which missing deeds correspond to different allocations and narrow the calibration. The policy question is not whether families should retain homes. It is whether property-tax inheritance rules reward retention when another allocation would generate greater net surplus, and how often that margin is relevant in practice.

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Appendix

A Data Construction Details

Source and scope. The CoreLogic Owner Transfer file compiles recorded deeds from county recorder offices in all fifty states and the District of Columbia. Our extract contains all record types—deeds of sale, quitclaims, intra-family conveyances, trust transfers, and foreclosure instruments. Coverage density is stable from 2007 onward; recording lags truncate the file around August 2024. All annual statistics use 2007–2023; the Proposition 19 cohort analyses use outcome windows that close by June 30, 2024.

Deduplication. Multi-parcel transactions and re-recorded corrections generate multiple rows per economic event. We keep one record per parcel-date pair, preferring arm's-length category records and, within category, the highest recorded consideration.

Classification. The taxonomy in Section 2 is implemented with the recorder-derived interfamily indicator, the arm's-length category code, recorded consideration, and string features of the buyer and seller name fields (surname equality; exact same-person matches on either buyer slot, which define the same-person retitle class; trust keywords; estate, executor, probate, and administrator keywords). Name fields are populated for the large majority of records (the seller surname is non-missing for 84 percent of interfamily events and the buyer surname for 66 percent); missing names can only push true family transfers into the "other interfamily" class, never into the market class, but imperfect full-name matching can also misclassify retitling. The same-surname series is a narrower proxy, not a formally established lower bound on true inter-person transfers.

Absentee recipients. A recipient is absentee when the buyer mailing ZIP on the deed differs from the property's situs ZIP, among records where both are observed (77 percent of events in 2007, rising to 98 percent by 2023). Time effects absorb common coverage changes; differential changes across states or transfer classes remain a measurement concern.

A.1 Recorded property types

We retain the Owner Transfer land-use and property-indicator codes as text. Detailed labels use five entries from a published CoreLogic crosswalk; remaining codes are retained

in the data without an asserted detailed label. The property indicator provides a broader cross-check: in this extract, indicator 11 accompanies both condominium and apartment land-use codes. It therefore cannot distinguish those assets by itself.

Table A1: Recorded property types and family deed flows, 2007–2023

Recorded type	Family deeds	Market deeds	Family share (%)	Family/market
Apartment	18,094	75,842	19.3	0.24
Condominium	1,457,215	8,283,796	15.0	0.18
Duplex	404,453	1,113,954	26.6	0.36
Single-family	15,057,327	50,442,379	23.0	0.30
Townhouse/rowhouse	323,061	2,167,932	13.0	0.15

Notes: Family deeds use the broad definition, excluding trust self-transfers. Shares divide family deeds by family plus market deeds. Partial published numeric crosswalk: 163 single-family, 102 townhouse/rowhouse, 112 condominium, 115 duplex, 106 apartment. Other codes remain unverified and are omitted from this exhibit, not the aggregate sample. Types are static recorded characteristics. Apartments are selectively covered: only 13% of source apartment rows pass the existing residential flag. Parcel deeds count neither dwelling units nor occupancy changes.

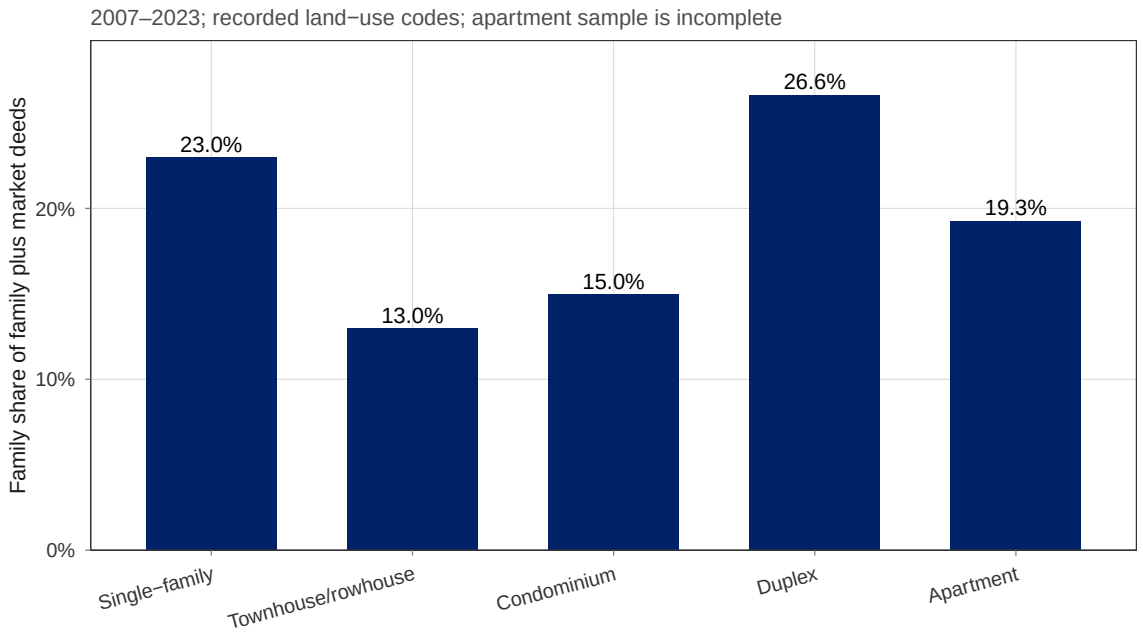


Figure A1: Family share of family-plus-market deeds by recorded type, 2007–2023. The broad family definition excludes trust self-transfers. Apartment coverage is incomplete.

Types are static recorded characteristics and may differ across records for a parcel. Counts are parcel-date deeds, not dwelling units. In the full source audit, only 13 percent of apartment-coded rows carry the residential flag required by the existing panel. Apart-

ment comparisons therefore do not describe the complete multifamily sector. We preserve the original residential restriction for this reproduction.

Type-specific five-year resale comparisons also differ. Among 2008–2018 cohorts, weighting broad-family and market events to the same market state/cohort distribution gives sale rates of 32.1 versus 26.4 percent for condominiums, 19.6 versus 22.0 percent for duplexes, and 24.0 versus 23.4 percent for single-family properties. These comparisons pool family classes, unlike the main-text same-surname benchmark, and do not identify an effect of inheriting an asset. They caution against a uniform interpretation of family deeds as lock-in.

Table A2: Ten-year retention by recorded property type

	Unsold at ten years (%)		Gap
Recorded type	Same-surname family	Market purchase	(pp)
Condominium	48.9	51.7	-2.8
Duplex	68.4	62.8	5.6
Single-family	61.8	58.4	3.4
Townhouse/rowhouse	54.3	52.3	2.0

Notes: Kaplan–Meier survival to the next market sale for 2008–2018 deed cohorts, censored June 2024, using the monthly exit construction of the aggregate analysis. These are pooled within-type comparisons; state/cohort composition, recipient selection, and occupancy histories are not controlled. Apartment results are omitted because of selective coverage.

The same-surname comparison at ten years also differs by recorded type. Kaplan–Meier estimates leave 68.4 percent of duplex family properties unsold versus 62.8 percent of market purchases, and 61.8 versus 58.4 percent for single-family properties. For condominiums, the ordering reverses: 48.9 percent of same-surname family properties remain unsold versus 51.7 percent of market purchases. These pooled comparisons do not hold state and cohort composition fixed, but the heterogeneity persists when the family class is restricted to the same-surname proxy.

An exploratory repetition of the transparent 2019-scaled deadline counterfactual yields anticipation excesses of 29.8 percent for single-family properties, 36.9 percent for condominiums, and 50.5 percent for duplexes. These comparisons have neither type-interaction inference nor direct measures of statutory eligibility; they are descriptive heterogeneity checks. Building type alone does not establish whether the property is a qualifying principal residence.

B Robustness Checks

Trust-inclusive volume DiD. Re-estimating the Proposition 19 volume difference-in-differences with trust self-transfers included in the family aggregate yields a CA × Post coefficient of −0.241 (s.e. 0.027), a 21 percent decline, confirming that the steady-state drop in family transfers is not an artifact of reclassification between direct family deeds and trust vehicles.

Synthetic-control in-space placebo. The main-text inference for the steady-state decline assigns the treatment to each donor state in turn and ranks California's post/pre mean squared prediction error ratio in the resulting distribution; California is second of fifty-one (Figure A2, placebo-rank $p = 0.039$).

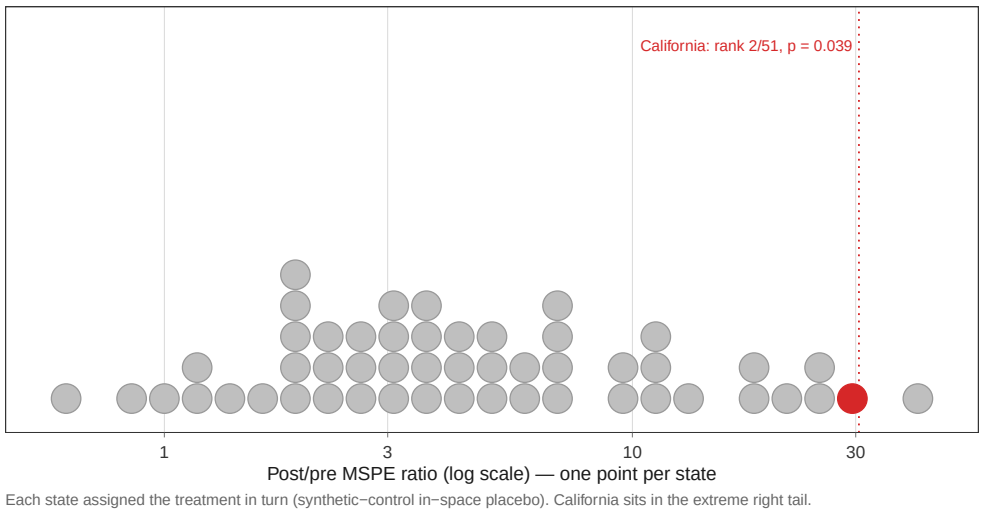


Figure A2: Synthetic-control in-space placebo: the post/pre mean squared prediction error ratio for each state assigned the treatment in turn. California (red) is second of fifty-one ($p = 0.039$).

SCM inference sensitivity. The saved Conley–Taber-style calculation gives $p = 0.167$ and a normalized-index gap interval of $[-0.626, 0.185]$. This interval includes zero and is in index-gap units, not percentage-change units. The population-adjusted calculation gives $p = 0.188$ under the same alternative procedure. These results qualify the sharper MSPE-ratio ranks; different statistics and assumptions need not give the same inference.

Permutation and pre-fit-adjusted ranks. As transparent alternatives to the synthetic control, each donor state is assigned the treatment in turn, each design is re-

estimated, and California's coefficient is ranked in the placebo distribution. For the annual family-volume DiD, California ranks 5th of 51 ($p = 0.098$); for the placebo-netted estimate, 9th ($p = 0.176$). The monthly counterfactual also reports ranks after scaling each state's policy-window gap by its own 2018m1–2020m10 RMSPE. Under the 2019 same-month baseline, California ranks 3rd of 51 in raw anticipation, February, and 2022–2023 gaps ($p = 0.059$), but 1st of 51 after pre-fit scaling in all three windows ($p = 0.020$). Using a 2018–2019 same-month baseline leaves the pre-fit-adjusted ranks unchanged. For the sold-within-24-months triple difference, $p = 0.67$; for the estate/trust-seller market-sale DiD, $p = 0.71$ (levels) and $p = 0.63$ (net of overall market).

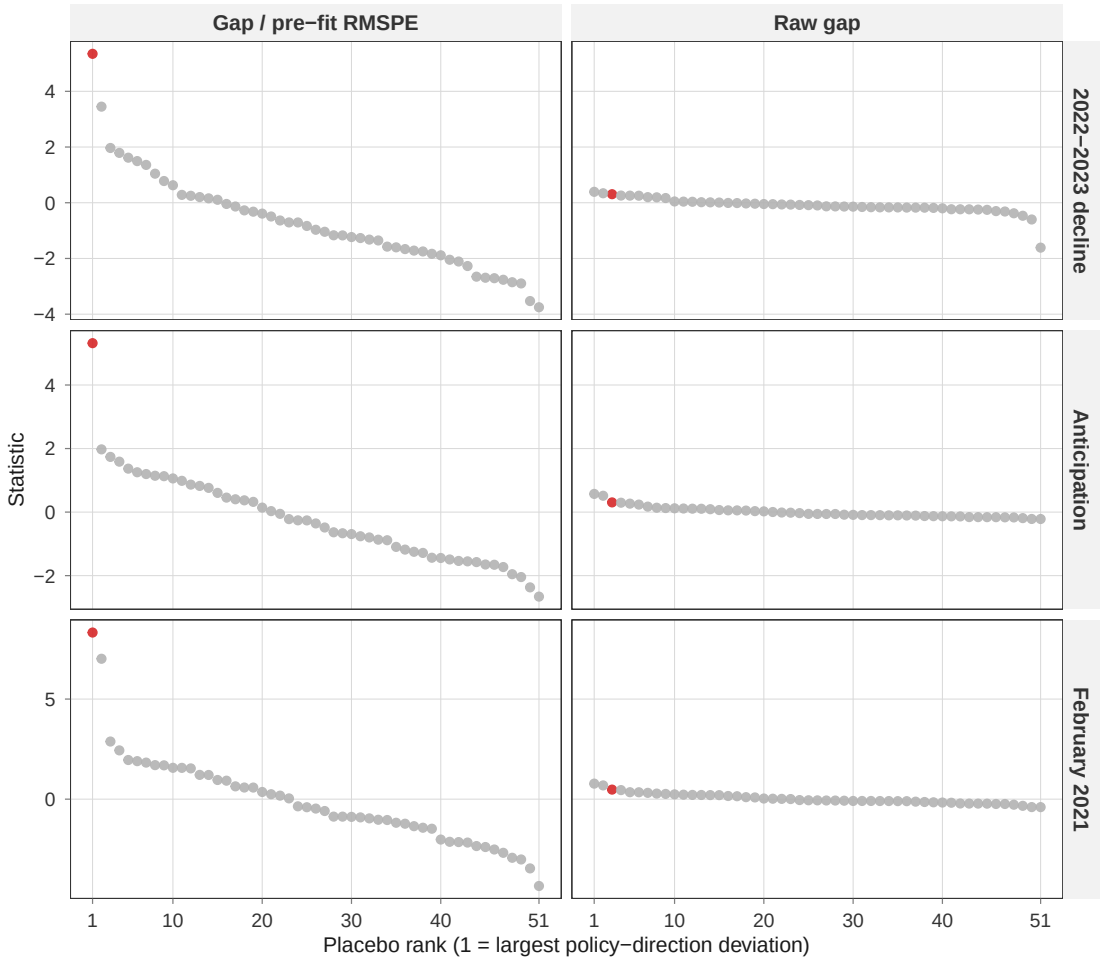


Figure A3: Placebo ranks for the monthly Proposition 19 counterfactual. Red points mark California; gray points assign the same policy window to each donor state in turn. Raw gaps rank California third in the baseline comparison, while gap divided by pre-period RMSPE ranks California first.

Selection: sold within 24 months by cohort. Figure A4 plots the conditional-margin series discussed in Section 4; it is relegated here because the triple difference is statistically unremarkable ($p = 0.67$) and the identified action is on the volume margin.

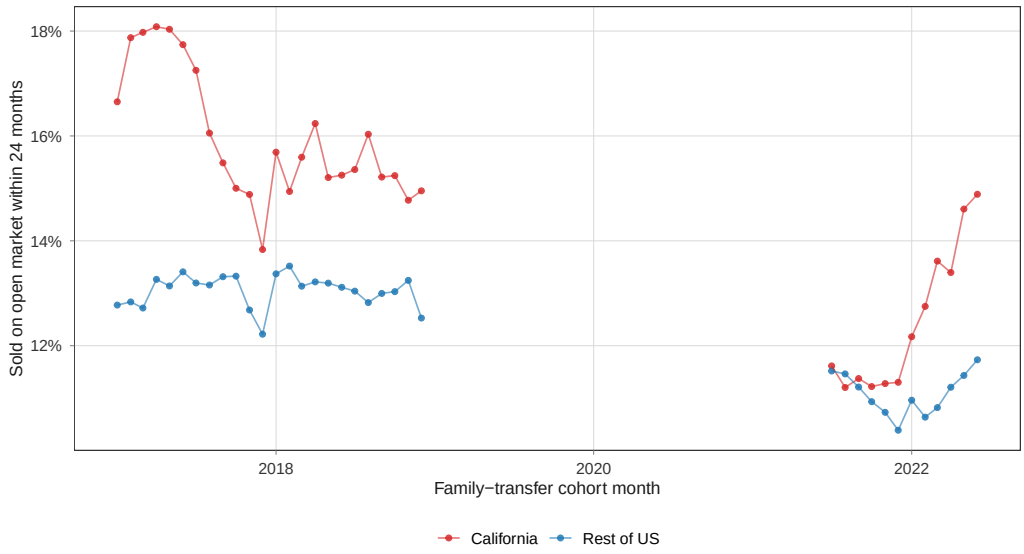


Figure A4: Share of family-transferred homes sold on the open market within 24 months, by cohort month. The plotted windows exclude the transition period around Proposition 19; lines are not connected across the omitted gap.

Raw difference-in-differences means. Table A3 reports the unadjusted means underlying the sold-within-24-months estimates.

Table A3: Sold on open market within 24 months: raw cohort means

Cohort group	Region	Pre (2017m1–2018m12)	Post (2021m7–2022m6)	Change
Family transfer	California	16.1% (n=470,163)	12.3% (n=247,417)	−3.8pp
Family transfer	Rest of US	13.1% (n=1,793,673)	11.1% (n=1,118,028)	−2.0pp
Market purchase	California	6.2% (n=788,627)	4.7% (n=421,807)	−1.5pp
Market purchase	Rest of US	6.9% (n=7,722,553)	6.5% (n=4,495,875)	−0.4pp

Notes: Non-corporate recipients only. Family transfer = same-surname, cross-surname interfamily, and estate/executor classes (same-person retitles excluded). The difference-in-differences for family transfers is −1.8pp; for market purchases, −1.1pp; the triple difference is −0.74pp (regression counterpart in Table 4; permutation $p = 0.67$).

C Welfare Accounting, Calibration, and Data Audit

C.1 Derivation and welfare boundary

The stationary benchmark fixes market prices, housing-service values, and tax rates. It assumes quasi-linear preferences, equal social weights on private households and taxpayers, and no unpriced externalities. If sale incurs real cost C_i, the private comparison is $(u_{hi} - t_i A_{0i})/r \geq p_i - C_i$. Substituting $p_i = u_{mi}/(r + t_i)$ gives $g_i \leq w_i$, where $g_i = u_{mi} - u_{hi} - rC_i$ and $w_i = t_i(p_i - A_{0i})$. The competitive-price and valuation assumptions are essential: observed selling prices alone do not identify $u_{mi} - u_{hi}$, and rental income cannot be added to the same tenant service value a second time. The finite-horizon totals retain this annual-equivalent cost convention; they do not separately estimate the timing of upfront moving or transaction costs. Short-run cash resource gains could differ if those costs arrive before the housing-service benefits.

With full removal of the advantage, the switching interval is $(0, w_i]$. Conditional uniformity gives $E[g_i \mid w_i, \text{switcher}] = w_i/2$. If a positive residual advantage w_{1i} remains, the interval is instead $(w_{1i}, w_{0i}]$, and under uniformity the average gain is $(w_{0i} + w_{1i})/2$, not half the removed advantage. The main calculation therefore restricts its interpretation to the fully removed margin; it does not apply the half-wedge formula to the surviving occupant exclusion.

For heterogeneous gains and durations the exact accounting object is

$$G_T = \sum_i g_i I_{iT},$$

where I_{iT} indicates that property i has switched allocation by T and its counterfactual loss would still persist. Replacing this sum by $\lambda w N_T$ requires a common mean gain among survivors or the corresponding independence approximation. Survival of legal ownership is not identical to persistence of a surplus gap. We use the former only as an explicitly imperfect proxy. The support envelope $0 \leq g_i \leq w_i$ is conditional on the model and the relevant switching population. It is not a data-identified welfare bound for the whole reform.

C.2 Episode construction and timing

Calendar dates are parsed from the eight-digit derived-sale-date field. Day 00 would be imputed to day 01; other impossible dates would be excluded and counted. In the current

Table A4: Incidence ledger for the inheritance provision

Component	Private or public incidence	Treatment in allocation calculation
Property-tax payments	Heir pays more; government receives more	Transfer; cancels with equal weights and budget offset
Efficient family retention	Attachment, insurance, and housing services preserved	Included in family value; no assumed loss
Alternative use	Buyers or tenants may obtain greater housing services	Count net resource gain only if allocation improves
Switching costs	Moving, transactions, administration, or displacement	Real costs enter the net gap; unmeasured costs limit interpretation
Asset-price changes	Seller/buyer capitalization effects	Held fixed; do not add capitalized taxes to their flow counterpart
Public spending and financing	Depends on how receipts are used	Not quantified; no claim about total policy welfare

148,711,427-event panel, neither case occurs. Explicit parsing reproduces the existing national hazard outputs. This exercise uses the derived sale date, not a separately verified legal succession date or recording date.

For California, an episode begins at the first qualifying family deed after the preceding market sale, or the first observed qualifying deed if there is no earlier observed sale. Episodes already beginning in 2007 are not restarted by a 2008 deed. Starts must fall in 2008–2018. Subsequent family deeds do not restart the clock. A later market sale ends the episode; a later family deed can begin a new, non-overlapping episode. Each parcel can therefore appear in more than one episode but cannot appear twice in the same outstanding stock. Broad-family and same-surname definitions are constructed separately. Missing pre-2007 history remains a limitation.

We restrict observed endpoints to June 30, 2024, estimate Kaplan–Meier survival using exact elapsed days and the associated risk sets, and evaluate it at annual ages using 365.25 days per year. Tied failures precede censoring at the same duration. Any collision created by date normalization is resolved to one parcel-date record, preferring a market sale; the current California audit finds none. The generated episode audit reports counts, repeated deeds, and the absence of overlapping intervals.

In the prospective accounting, cohorts enter at the beginning of each year 2022–2031, with $S(0) = 1$. Beginning-year stocks approximate property-years within each year; annual gains are discounted as year-end flows to the beginning of 2022. A five-year delay

gives no gains in 2022–2026, and a ten-year delay gives no gains within this horizon. The real-dollar base is 2023 throughout; the discount-date origin is a separate choice. We do not claim that uncounted benefits after 2031 vanish. Nor do we count a pre-reform, grandfathered stock as a gain caused by the reform.

The factor α incorporates unique-property conversion and changes in allocation, rather than mechanically dividing missing deeds by the national deeds-per-parcel ratio. Prospective cohorts are interpreted as distinct new property episodes; aggregate deed estimates cannot verify that assumption. The constant annual response and its attribution to the inheritance provision are maintained scenarios, not identified steady-state properties.

C.3 Valuation feasibility and external inputs

Table A5: California valuation audit

Check	Result
Property rows	8,848,364
Distinct nonmissing parcel IDs	8,492,816
Rows missing parcel ID	355,548
Market-value field observed (%)	0.0
Tax-year 2008–2009 rows (% of all rows)	95.9
Matched, dated 2017–2020 events with stale tax year (%)	100.0
Median event minus tax-year gap (years)	10

Notes: Stale means more than one year before the event. The alignment audit uses California family events and the latest available tax year per parcel. These fields do not identify contemporaneous pre-reform market values or assessment gaps. Full field, key, missingness, and vintage diagnostics are saved by the welfare script.

The California file contains assessed and calculated values, but its market-value field (`market_total_value`) and property-tax-rate field are missing. Tax-year coverage is concentrated in 2008–2009. Current file availability does not make its values contemporaneous with the reform. The calculated value's source field is audited, but without validated field definitions and transaction-time alignment we do not interpret it as an independent market estimate. Missing parcel IDs are counted separately; the valid IDs have no duplicate rows. The latest available tax year is used only for the vintage diagnostic, not to fabricate an assessment history. Consequently no parcel-level historical wedge is reported as measured.

The external illustration instead uses the LAO's \$3,000–\$4,000 saving for a long-held Los Angeles home. It is neither a statewide average nor the wedge of the policy-responsive properties. The source does not specify a dollar year; we assume its 2017 publication year.

Annual CPI-U values are 245.120 in 2017 and 304.702 in 2023, with the ratio used for conversion (U.S. Bureau of Labor Statistics, 2026). The $3,500 midpoint is an assumption. The LAO's separate 60,000–80,000 annual property count includes administrative eligibility and population differences and does not validate the deed response.

C.4 Sensitivity and statistical uncertainty

Table A6: Welfare sensitivity around an illustrative scenario

Persistence	Wedge ($)	λ	Hazard scale	Discount (%)	PV ($m)
Broad	2,175	0.50	1.0	3	328.7
Broad	3,729	0.50	1.0	3	563.5
Broad	4,351	0.25	1.0	3	328.7
Broad	4,351	0.50	0.5	3	693.7
Broad	4,351	0.50	1.0	0	847.4
Broad	4,351	0.50	1.0	3	657.4
Broad	4,351	0.50	1.0	5	557.8
Broad	4,351	0.50	2.0	3	594.1
Broad	4,351	0.75	1.0	3	986.1
Broad	4,972	0.50	1.0	3	751.3
Broad	8,702	0.50	1.0	3	1,314.8
Same surname	4,351	0.50	1.0	3	663.8

Notes: Every row assumes $\alpha = 0.5$ and a five-year delay; neither assumption is estimated. Dollar values are 2023 dollars. Hazard scale k replaces survival $S(a)$ with $S(a)^k$. All 4,050 joint scenarios are saved in the machine-readable output; this table changes one input at a time.

The full grid comprises 4,050 combinations: two episode definitions, five conversion factors, three delays, five wedges, three valuation-gap fractions, three hazard scales, and three discount rates. The wedge menu is $1,750, $3,000, $3,500, $4,000, and $7,000 in assumed 2017 dollars, all converted to 2023 dollars. The endpoints outside the LAO range are stress cases. Changing cumulative hazards by a factor k uses $S_k(a) = S(a)^k$. Discount rates are 0, 3, and 5 percent. The main table varies conversion and delay; the appendix changes other inputs around an illustrative, explicitly unpreferred $\alpha = 0.5$, five-year-delay case.

These calculations condition on the net-DiD reference flow. They do not turn uncertainty about that flow into statistical confidence bands for welfare. Its placebo-rank $p = 0.176$ and the SCM specifications' different inference remain material. A nonpositive true allocation response, title-only responses, a sufficiently long delay, or no net resource gap can eliminate the calculated gains. Correlation between wedges and persis-

tence, changing future tax liabilities, endogenous rents and prices, heterogeneous financing constraints, and real public-service effects are outside this calculation.